

llmXive Automated Review of LiveEdit: Towards Real-Time Diffusion-Based Streaming Video Editing

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: [qwen.qwen3.5-122b](#)

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and their alignment with the provided evidence.

Latency Inconsistency There is a direct mathematical contradiction regarding the reported inference speed. The Abstract and Introduction claim a throughput of **12.66 FPS** (approximately 79ms per frame). However, Table `tab:three_stages_ablation` (referenced in `sec/4_experiment.tex`) lists the latency for Stage 3 as **7.89s** for **81 frames**. Calculation: $81 \text{ frames} / 7.89 \text{ s} \approx 10.27 \text{ FPS}$ (or $7.89 \text{ s} / 81 \approx 97.4 \text{ ms/frame}$). The claimed 12.66 FPS corresponds to $\sim 79\text{ms/frame}$. The discrepancy between 79ms and 97ms is significant ($\sim 23\%$). The authors must clarify which metric is correct and ensure the text and table align. If 12.66 FPS is the target, the table’s latency or frame count is incorrect.

Causal Logic in Mask Cache In `sec/3_method.tex`, the AR-oriented Mask Cache is described as deriving the mask for chunk k from the edited output of chunk $k - 1$ (z_{edit}^{k-1}). The text states: “we derive its spatial editing mask from the generation trajectory of the preceding chunk.” In a real-time streaming scenario, if the system is processing chunk k *while* chunk $k - 1$ is being edited, the final edited latent z_{edit}^{k-1} might not be fully available or stable before the decision to cache for chunk k is made. If the mask relies on the *final* edited result of the previous chunk, this introduces a dependency that could violate strict real-time causality or introduce a one-chunk delay not explicitly accounted for in the “real-time” claim. The authors should clarify if the mask is generated from the *source* of $k - 1$ (which is available) or the *edited* output, and how this impacts the causal flow.

Support for Feature Similarity Claims The text in `sec/4_experiment.tex` claims that “SA tokens maintain exceptionally high temporal redundancy” and cites `fig:cosine` as validation. However, the text does not provide the specific quantitative values (e.g., “mean cosine similarity of 0.95”) observed in the figure. While the figure likely displays this, the textual claim “exceptionally high” is subjective without the supporting numbers in the prose. For a rigorous claim, the text should explicitly state the observed similarity metric values to substantiate the decision to cache Self-Attention layers.

Action items.

- **[writing]** Latency figures are inconsistent: Abstract claims 12.66 FPS ($\sim 79\text{ms}$), but Table 3 (`sec/4_experiment.tex`) shows 7.89s for 81 frames ($\sim 97\text{ms}$). Clarify the exact frame count and time used for the FPS claim.
- **[science]** In `sec/3_method.tex`, the mask for chunk k relies on the edited output of chunk $k-1$. Clarify if this edited output is available in real-time without introducing a causal delay or dependency violation.
- **[writing]** The claim of ‘exceptionally high’ temporal redundancy for SA tokens in `sec/4_experiment.tex` lacks specific quantitative values from Fig. 5. Include the mean cosine similarity values to support this assertion.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 effectively communicates the paper’s core contribution by visually contrasting three video editing paradigms. The layout is clear, the check/cross indicators are unambiguous, and the visual examples directly support the claims made in the caption regarding accuracy and efficiency.

Figure 2

Figure 2 contains a significant contradiction between the right panel’s visualization (triangular mask) and its caption (claiming uniform spread). Additionally, the y-axis label ‘Key’ is poorly formatted and partially illegible.

Figure 3

The figure provides a clear overview of the three-stage framework and the AR-oriented Mask Cache. However, the title for Stage 2 is truncated at the right edge, and there is a minor inconsistency in the naming of the bidirectional model between the diagram and the legend.

Figure 4

The figure effectively visualizes the mask generation process, but the top-right histogram mislabels its x-axis as ‘Pixel Change Rate’ while displaying ‘Temporal IoU’ statistics, and the difference matrix row is missing a color scale legend.

Figure 5

The figure effectively demonstrates the qualitative superiority of the proposed method over baselines, but the layout is cluttered with rotated row labels embedded in the grid and instruction text overlaid on the source images.

Figure 6

Figure 6 effectively visualizes the impact of different cache locations on video editing quality. The grid layout clearly compares the input against three ablation settings, and the caption provides the necessary context regarding the editing instruction.

Figure 7

The figure clearly visualizes the distribution of cosine similarity for two mechanisms, but the caption is grammatically incorrect and lacks specificity regarding what the two distributions represent.

Figure 8

The figure presents user study results clearly, but the caption’s claim that lines represent ‘top-3 selections’ is inaccurate as the lines track specific rank categories (‘Best’/‘Second’) rather than a cumulative sum, and the ‘Best’ line value in the top chart conflicts with the stacked bar data.

Figure 9

Figure 9 effectively demonstrates the versatility of the LiveEdit system through a series of qualitative examples. The layout clearly distinguishes between user inputs and the resulting edits, with descriptive captions for each row explaining the specific transformation applied.

Figure 10

Figure 10 effectively demonstrates the versatility of the LiveEdit system through a series of qualitative examples. The layout is clear, with distinct ‘User input’ and ‘LiveEdit’ columns, and the text descriptions below each row provide necessary context for the edits performed.

Figure 11

Figure 11 effectively demonstrates the versatility of the LiveEdit system through seven distinct qualitative examples, ranging from material changes (chess pieces, statues) to lighting and texture modifications. The layout is clear, with ‘User input’ images on the left and the resulting ‘LiveEdit’ outputs on the right, accompanied by descriptive text explaining the specific transformation for each case.

Figure 12

Figure 12 presents a qualitative comparison of video editing methods but suffers from a generic caption that omits the specific task details and uses undefined abbreviations for the baseline methods, reducing its standalone clarity.

Action items.

- **[science]** Figure 2: The right panel (‘Directly Truncate’) shows a triangular attention mask where attention is restricted to the causal past (lower triangle), yet the caption claims it ‘forces attention to spread uniformly across all historical frames.’ The visualization contradicts the text description; the attention is sparse (triangular), not uniform across the full history.
- **[writing]** Figure 2: The y-axis label ‘Key’ is rotated 90 degrees and partially cut off at the bottom (e.g., the ‘y’ in ‘Key’ is missing or obscured), and the tick labels on the right plot’s y-axis are misaligned or missing for some values.

- **[writing]** Figure 3: The title for Stage 2 is cut off on the right edge ('...Causal Initial'), making the full stage name illegible.
- **[writing]** Figure 3: The legend in Stage 2 defines 'Bid. Model' but the corresponding block in Stage 1 is labeled 'Bidirect. DiT', creating inconsistent terminology.
- **[science]** Figure 4: The 'Temporal IoU Distribution' histogram shows a mean of 0.016%, yet the x-axis is labeled 'Pixel Change Rate (%)'. IoU (Intersection over Union) and Pixel Change Rate are distinct metrics; the axis label should match the metric name or the caption should clarify the relationship.
- **[writing]** Figure 4: The 'Difference Matrix' row lacks a colorbar or legend to define the mapping between the heatmap colors (blue to red) and the magnitude of pixel differences.
- **[writing]** Figure 5: The row labels 'LucyEdit', 'InsV2V', 'Stream.V2', and 'Ours' are rotated 90 degrees and placed inside the image grid, which is visually cluttered and unconventional for a comparison table.
- **[writing]** Figure 5: The instruction text is placed directly over the image content in the top row, reducing readability and obscuring parts of the source video.
- **[writing]** Figure 7: The caption states the plot shows 'token cosine similarity between consecutive denoising step' (singular), but the figure displays two distinct distributions (Self-Attn and FFN) without clarifying in the text that these represent different attention mechanisms or layers.
- **[writing]** Figure 7: The caption contains a grammatical error ('denoising step' should be 'denoising steps').
- **[science]** Figure 8: The caption states 'line plots indicate the proportion of top-3 selections,' but the line plots track the 'Best' category (dark blue) and the 'Second' category (medium blue) separately, not a combined top-3 sum. Additionally, the 'Best' line for 'Ours' reaches 100% in the top chart, which contradicts the stacked bar showing ~96% 'Best' and ~4% 'Second' (implying <100% 'Best').
- **[writing]** Figure 8: The legend defines 'Best', 'Second', 'Third', and 'Others', but the line plots only visualize 'Best' and 'Second' (or a mix of them), leaving the 'Third' and 'Others' categories unrepresented in the line plot overlay, creating ambiguity about what the lines specifically represent.
- **[science]** Figure 12: The 'User Input' row displays four distinct video frames, but the editing instruction is a single static text prompt. This implies the input is a video, yet the instruction does not specify how the edit should apply across the temporal dimension (e.g., to all frames or a specific segment), making the evaluation setup ambiguous.
- **[writing]** Figure 12: The caption 'More comparison between baseline and our LiveEdit' is generic and fails to describe the specific editing task shown (changing goose feathers to blue), forcing the reader to rely on the image text which is not part of the formal caption.

- **[writing]** Figure 12: The row labels (e.g., ‘InsV2V’, ‘LucyEdit’, ‘Stream.’) are abbreviations that are not defined in the caption or the figure itself, requiring the reader to guess the full method names.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript relies heavily on domain-specific acronyms and jargon that are not defined at their first occurrence, creating a barrier for non-specialist readers. Specifically, the terms ‘AR’ (Augmented Reality), ‘DiT’ (Diffusion Transformer), ‘DMD’ (Distribution Matching Distillation), ‘FFN’ (Feed-Forward Network), ‘NFEs’ (Network Function Evaluations), and ‘CFG’ (Classifier-Free Guidance) are introduced in the Abstract, Introduction, or early Method sections without expansion. For instance, ‘AR-oriented’ appears in the Abstract before ‘AR’ is defined, and ‘DiT’ is used in the Introduction without spelling out ‘Diffusion Transformer’.

Furthermore, the term ‘chunk’ is used repeatedly (e.g., ‘chunk-by-chunk’, ‘chunk-wise’) to describe the temporal processing unit, yet the paper never explicitly defines what constitutes a ‘chunk’ (e.g., number of frames, duration) until perhaps implicitly in the implementation details, leaving the reader to guess the granularity of the operation. The phrase ‘naively truncated’ in the Introduction is also stylistically questionable; ‘simply’ or ‘directly’ would be more neutral and precise.

To improve accessibility, the authors should ensure every acronym is spelled out upon its first appearance in the text. Additionally, the term ‘chunk’ should be explicitly defined with its specific parameters (e.g., ‘a temporal chunk of 3 frames’) in the Method section where it is first introduced as a core concept. Replacing subjective adjectives like ‘naively’ with neutral technical descriptors will also enhance the professional tone of the paper.

Action items.

- **[writing]** The manuscript relies heavily on domain-specific acronyms and jargon that are not defined at their first occurrence, creating a barrier for non-specialist readers. Specifically, the terms ‘AR’ (Augmented Reality), ‘DiT’ (Diffusion Transformer), ‘DMD’ (Distribution Matching Distillation), ‘FFN’ (Feed-Forward Network), ‘NFEs’ (Network Function Evaluations), and ‘CFG’ (Classifier-Free Guidance) are introduced in the Abstract, Introduction, or early Method sections without expansion. For instance, ‘

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a coherent high-level argument for streaming video editing but contains logical gaps in the causal mechanisms of its core contributions.

First, the **AR-oriented Mask Cache** (Section 3.3) suffers from a causality error. The mask M^k for the current chunk k is derived from the difference between the edited and source latents of the *previous* chunk ($k - 1$). The text then uses this mask to route computation for chunk k . This logic fails if the editing region moves or changes between chunks (e.g., a moving object being edited). The state of chunk $k - 1$ does not causally determine the editing activity in chunk k . Without a mechanism to predict or detect the edit location in k before processing it, the mask will be misaligned, leading to artifacts or missed edits. The paper must explain how the mask for k is generated without access to the ground truth of k .

Second, the **pruning logic** in Section 4.1 is contradictory. The authors state the threshold τ is set to “explicitly prune 70% of the redundant spatial tokens.” This implies a fixed ratio regardless of content. However, the method claims “strict background preservation.” If an edit covers a large portion of the frame, a fixed 70% prune ratio might force the system to treat active editing regions as static background, violating the preservation claim. The logic of a fixed ratio is incompatible with variable edit sizes unless “70%” refers to a dataset average, which is not clarified.

Finally, the **ablation results** in Table 2 (Section 4.3) show Dynamic Degree (DD) collapsing to 0.017 when caching FFN layers. The authors attribute this to “high-frequency spatial information” loss (blurring). However, DD measures motion consistency; a value near zero suggests the video became static or motion was lost entirely, not just blurred. The causal link between FFN caching and the specific metric collapse (motion freezing vs. blurring) is not logically explained.

Action items.

- **[science]** Section 3.3 claims the mask for chunk k is derived from chunk $k-1$. Logically, an edit moving in chunk k cannot be masked correctly using $k-1$ ’s data. Explain how the mask for k is determined causally without ground truth.
- **[writing]** Section 4.1 states a fixed 70% pruning ratio. This contradicts ‘strict preservation’ if the edit region varies. Clarify if 70% is an average or if the logic holds for large edits.
- **[science]** Table 2 shows Dynamic Degree drops to 0.017 with FFN caching. The text blames ‘blurring,’ but near-zero DD implies static video. Explain if the metric collapse is due to motion freezing.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several strong claims regarding the performance and capabilities of the LiveEdit framework that exceed the evidence provided in the text and tables.

First, the authors repeatedly use absolute language such as “zero visual degradation” (Abstract, Sec 3.3, Sec 5) and “strictly guaranteeing absolute visual consistency” (Sec 3.3). This is an over-claim. The quantitative results in Table 3 (Ablation Study) explicitly show that enabling the cache results in a decrease in Aesthetic Quality (from 0.584 to 0.581) and Imaging Quality (from 0.720 to 0.708) compared to the “W/o Cache” baseline. While these drops may be small, they are non-zero. Claiming “zero degradation” contradicts the provided data. The text should be softened to reflect that the degradation is minimal or imperceptible to human evaluators, rather than mathematically non-existent.

Second, the claim in Section 4.3 that the unidirectional streaming approach “strictly outperforms” bidirectional offline architectures in Text Alignment is slightly misleading. While the final model (0.270) does outperform InsV2V (0.259), the “W/o Cache” version (0.265) also outperforms it, but the margin is narrow. More importantly, the claim implies a general superiority over the entire class of offline models, yet the table shows the method is competitive but not universally dominant across all metrics against all baselines (e.g., StreamDiffusionV2 has a higher Motion Smoothness score). The phrasing should be more precise, such as “achieves state-of-the-art performance among streaming methods and competitive results against offline baselines.”

Finally, the assertion that the Mask Cache mechanism “guarantees” strict background preservation (Sec 3.3) relies on a heuristic threshold τ to prune 70% of tokens. While the ablation study supports the efficacy of this approach, the use of “guarantee” suggests a theoretical bound or perfect execution that is not demonstrated. The method is empirical; if the L_2 distance threshold fails to capture subtle changes in a specific scene, artifacts could theoretically occur. The language should be adjusted to “effectively preserves” or “maintains high-fidelity consistency” to accurately reflect the empirical nature of the results.

Action items.

- **[writing]** The claim of ‘zero visual degradation’ in unedited regions (Sec 3.3, Sec 5) is an absolute over-claim. Table 3 shows a drop in Aesthetic Quality (0.584 to 0.581) and Imaging Quality (0.720 to 0.708) when the cache is enabled. The text must be revised to reflect that degradation is ‘negligible’ rather than non-existent.
- **[writing]** The assertion that the method ‘strictly outperforms’ bidirectional offline models in Text Alignment (Sec 4.3) is not fully supported by the data. While the final score (0.270) beats InsV2V (0.259), the claim implies universal superiority over the class, which is not demonstrated against all baselines. The phrasing should be qualified to ‘outperforms specific strong baselines’ or ‘achieves competitive SOTA’.
- **[writing]** The claim that the AR-oriented Mask Cache ‘guarantees’ absolute visual consistency (Sec 3.3) is an over-interpretation of the L_2 distance heuristic. The method relies on a threshold τ to prune 70% of tokens; if the threshold is not perfectly tuned for every scene, artifacts could occur. The word ‘guarantees’ should be replaced with ‘significantly preserves’ to align

with the empirical nature of the evaluation.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript presents a technically advanced framework for real-time video editing but lacks sufficient discussion on safety and ethical implications.

First, regarding human subjects research, Section X_suppl describes a user study involving 20 volunteers. The text details the metrics and results but fails to mention ethical compliance. Standard practice requires explicit confirmation of Institutional Review Board (IRB) approval or an exemption determination, along with a statement confirming that informed consent was obtained from all participants. The manuscript should be updated to include these details to ensure the research adheres to ethical standards for human-subject studies.

Second, the training data provenance is insufficiently addressed. The authors state in Section 4.1 that the dataset comprises “20K high-quality video-video pairs, which are carefully filtered from the large-scale Ditto-1M dataset.” There is no discussion regarding the licensing, consent, or privacy status of the source videos within Ditto-1M. Given the potential for training data to contain personally identifiable information (PII) or copyrighted material, the authors must clarify the ethical sourcing and legal compliance of this dataset to prevent potential copyright infringement or privacy violations.

Third, the paper highlights significant dual-use risks. The ability to perform high-fidelity, real-time video editing with strict background preservation (12.66 FPS) could be misused for generating deepfakes, disinformation, or unauthorized surveillance tools. The current manuscript focuses almost exclusively on performance metrics and technical novelty. A dedicated section on “Ethical Considerations” or “Societal Impact” is necessary to acknowledge these risks. The authors should discuss potential misuse scenarios and propose mitigation strategies, such as the integration of invisible watermarks, detection mechanisms, or strict usage policies for the released code and models.

Finally, while the paper mentions a “dedicated benchmark,” it does not specify if this benchmark includes safety-critical scenarios or if it was curated to avoid sensitive content (e.g., faces of private individuals). Clarifying the composition of the benchmark would further strengthen the ethical rigor of the evaluation.

Action items.

- **[writing]** The manuscript presents a technically advanced framework for real-time video editing but lacks sufficient discussion on safety and ethical implications. First, regarding human subjects research, Section X_suppl describes a user study involving 20 volunteers. The text details the metrics and results but fails to mention ethical compliance. Standard practice requires explicit confirmation of Institutional Review Board (IRB) approval or an

exemption determination, along with a statement confirming

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The scientific evidence supporting the core claims of real-time performance and quality preservation requires clarification regarding metric consistency and statistical rigor.

First, there is a discrepancy in the reported inference speed. The Abstract claims “12.66 FPS,” while the Implementation Details section states an “ultra-low latency of 79ms per-frame” (which mathematically aligns with ~ 12.66 FPS). However, the ablation study in `tab/ablation-arch.tex` reports a total latency of “7.89s for 81 frames,” which calculates to approximately 10.27 FPS. The authors must explicitly define the hardware configuration, input resolution, and specific video length used for the 12.66 FPS claim to resolve this inconsistency and ensure the “real-time” claim is robustly supported.

Second, the evidence for the “indistinguishable” quality of the AR-oriented Mask Cache is weakened by the quantitative results in `tab/ablation_cache.tex`. While the authors claim the cache introduces “no degradation,” the table shows a measurable drop in Imaging Quality (0.720 to 0.708) and Aesthetic Quality (0.584 to 0.581) when the cache is active compared to the “W/o Cache” baseline. The text in the “Effectiveness of AR-oriented Mask Cache” section asserts that the SA-cache version “successfully modify[s] the target object with high fidelity,” but the data suggests a trade-off. The authors should either adjust the qualitative claim to reflect a “negligible” rather than “indistinguishable” impact or provide a statistical test (e.g., paired t-test) demonstrating that these metric differences are not significant.

Finally, the user study results in the Supplementary Material (`sec/X_suppl.tex`) rely on a small sample size (20 volunteers) to support the strong claim of “100.0% top-3 preference” for Instruction Consistency. Without reporting the total number of video comparisons made or a statistical significance test (e.g., a binomial test against a random baseline), this result is vulnerable to overfitting or selection bias. The authors should provide the raw data counts and statistical validation to substantiate the claim of overwhelming superiority.

Action items.

- **[science]** The claim of 12.66 FPS (Abstract) conflicts with the reported 79ms/frame latency (Implementation Details). $1/0.079s \approx 12.66$ FPS, but the ablation table (`tab/ablation-arch.tex`) lists 7.89s latency for 81 frames, which is ~ 10.2 FPS. Clarify the exact benchmark conditions (resolution, hardware, chunk size) for the FPS claim to ensure reproducibility.
- **[science]** The user study (`sec/X_suppl.tex`) reports 100% top-3 preference for Instruction Consistency with only 20 volunteers. Without reporting the total number of video pairs evaluated or the statistical significance (e.g., binomial test p-value), this result risks being

an artifact of small sample size or selection bias. Provide the raw counts and statistical validation.

- **[science]** The AR-oriented Mask Cache claims to prune 70% of tokens (Implementation Details) while maintaining ‘indistinguishable’ quality. The ablation table (tab/ablation_cache.tex) shows a drop in Imaging Quality (0.720 -> 0.708) and Aesthetic Quality (0.584 -> 0.581) when the cache is enabled. The authors must reconcile the claim of ‘indistinguishable’ quality with these measurable metric degradations.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical rigor of the evaluation section requires strengthening to support the strong claims of state-of-the-art performance and real-time efficacy.

First, the **User Study** (Section “User Study” in `sec/X_suppl.tex`) relies on percentages derived from a sample size of $N = 20$ volunteers. While the reported preference rates (e.g., 100.0% for Instruction Consistency) are striking, the absence of statistical significance testing (such as a binomial test against a null hypothesis of random choice or a chi-square test for independence) makes it difficult to rule out chance. Furthermore, no confidence intervals (e.g., 95% CI) are provided for these proportions, which is standard practice for human evaluation studies to indicate the precision of the estimate.

Second, the **Quantitative Comparisons** (Tables `tab/baseline.tex` and `tab/ablation_cache.tex`) present point estimates for metrics like Text Alignment and Background Consistency to three decimal places. Diffusion-based generation is inherently stochastic; therefore, reporting a single mean value without the standard deviation (std) or confidence intervals obscures the stability of the method. For instance, the difference between “Ours (W/ Cache)” (TA=0.270) and “Ours (W/o Cache)” (TA=0.265) appears marginal. Without variance data, it is impossible to determine if this improvement is statistically significant or within the noise floor of the evaluation metric.

Third, regarding the **AR-oriented Mask Cache** (Section 4.1, `sec/4_experiment.tex`), the authors state that the threshold τ is dynamically calculated to “explicitly prune 70% of the redundant spatial tokens.” It is unclear if this 70% figure is a target hyperparameter, a mean achieved across the dataset, or a fixed constant. If it is a mean, the standard deviation of the pruning ratio across the 120 test pairs should be reported to demonstrate the consistency of the cache’s behavior. If it is a fixed target, the text should clarify that the threshold τ is adjusted per frame to meet this specific sparsity constraint, rather than being a static value.

Finally, the **Ablation Study** (Table `tab/ablation-arch.tex`) compares latency and quality across stages. While the latency drop is dramatic, the quality metrics (Text Alignment, Image Quality) show minor fluctuations (e.g., 0.268 vs 0.264). Without statistical testing or error bars,

the claim that Stage 3 “maintains high-quality editing performance” despite the aggressive distillation is not fully statistically substantiated. The authors should add variance metrics to all quantitative tables and perform significance tests on the user study results.

Action items.

- **[science]** The user study (Sec. X_suppl) reports percentages (e.g., 100.0%, 75.0%) from 20 volunteers but lacks statistical significance testing (e.g., binomial test, chi-square) or confidence intervals to validate that the observed superiority over baselines is not due to chance.
- **[science]** Quantitative results in Tab. 1 and Tab. 2 report metrics to three decimal places without standard deviations or confidence intervals. Given the stochastic nature of diffusion models, variance estimates are required to assess the reliability of the reported SOTA claims.
- **[science]** The claim of pruning ‘70% of redundant spatial tokens’ (Sec. 4.1) is presented as a fixed empirical outcome. The manuscript should clarify if this threshold was optimized via cross-validation or if it represents a mean across the test set, and provide the variance of this pruning ratio.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript presents a clear and generally well-structured narrative, effectively communicating the motivation, methodology, and results of the LiveEdit framework. The logical flow from identifying bottlenecks in streaming video editing to proposing a three-stage distillation pipeline and an AR-oriented Mask Cache is coherent and easy to follow. Technical terms are generally well-defined upon first use, and the distinction between the proposed method and existing baselines is articulated with precision.

However, the text contains several minor grammatical errors and punctuation issues that detract from the overall polish and professionalism of the paper. In the Introduction (sec/1_introduction.tex), the first contribution bullet point lacks a space before the parenthesis in “latency(12.66 FPS)”. In the Experiment section (sec/4_experiment.tex), the description of Stage 3 begins with a sentence fragment (“deliberately bypassing..”) that disrupts the grammatical flow. Additionally, a subject-verb agreement error appears in the Method section (sec/3_method.tex) where “process demonstrate” should be “process demonstrates”.

The supplementary material (sec/X_suppl.tex) exhibits slightly more carelessness in proofreading. The User Study section contains a sentence ending with a comma followed immediately by a period (“accurately, .”), and the caption for the user study figure ends with a double period (“dimensions.”). These errors, while minor, suggest a lack of final proofreading pass. Addressing these specific mechanical issues will significantly improve the readability and presentation quality

of the manuscript.

Action items.

- **[writing]** In sec/1_introduction.tex, the first contribution bullet point contains a missing space before the parenthesis: ‘latency(12.66 FPS)’ should be ‘latency (12.66 FPS)’.
- **[writing]** In sec/4_experiment.tex, the Stage 3 bullet point begins with a sentence fragment: ‘deliberately bypassing the computationally expensive ODE initialization.’ This should be integrated into the preceding sentence or rephrased as a complete sentence.
- **[writing]** In sec/4_experiment.tex, the sentence ‘our mask generation process demonstrate extremely high structural stability’ contains a subject-verb agreement error; ‘process’ is singular and should be followed by ‘demonstrates’.
- **[writing]** In sec/X_suppl.tex, the User Study section ends a sentence with a comma and a period: ‘...accurately,.’ This punctuation error must be corrected.
- **[writing]** In sec/X_suppl.tex, the caption for Fig. ef{fig:user_study} ends with a double period: ‘...dimensions.’ Remove the extra period.